Question Answering

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Based on slides by Pasquale Minervini

Coursework

- The coursework has been pre-released on the Opencourse website (https://opencourse.inf.ed.ac.uk/atnlp/assessment).
- Please note that it is officially released on 9/2/2026 – prerelease more time
- Read the instructions through the website, the PDF, and the README.md files early to know how to prepare and manage your time.
- If you have any questions, please post them on Piazza.
- In addition, you will have to submit the partnering form here: https://edin.ac/4jKANHn . Please read the instructions on the form carefully, as *only* one partner for a given pair needs to submit it. The deadline for submitting the form is 26/1/2026 (Monday) at 5pm - so that leaves you more than a week to do that.
- Once we set up the partners, we hope to obtain the necessary ELM credits for solving Part 1 of the coursework quickly.

Outline

Aim: Understand how machines learn to read texts and answer questions

- Introduction and concept
- Reading Comprehension
- Open Domain QA
- Summary

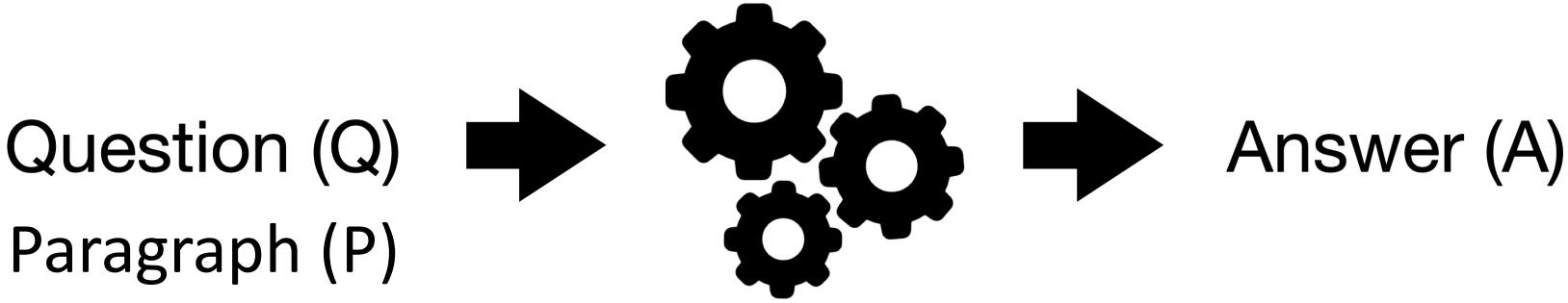
Readings:

- Kwiatkowski et al. (2019). Natural Questions: A Benchmark for Question Answering Research TACL (Sections 1-3)

- Izacard and Grave (2021). Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering, EACL (Sections 1-3)

Introduction

What is Question Answering?



In question answering, we aim to build systems that can **automatically** answer questions posed by humans in **natural language**.

What type of questions?

Factoid vs. Non-Factoid; Open-Domain vs. Closed-Domain; Simple vs. Compositional; Natural vs. Cloze-style; …

What type of answers?

Short text; Paragraph; List; Yes/No; Multiple Choice

Applications

AI Mode All Short videos Forums Images More

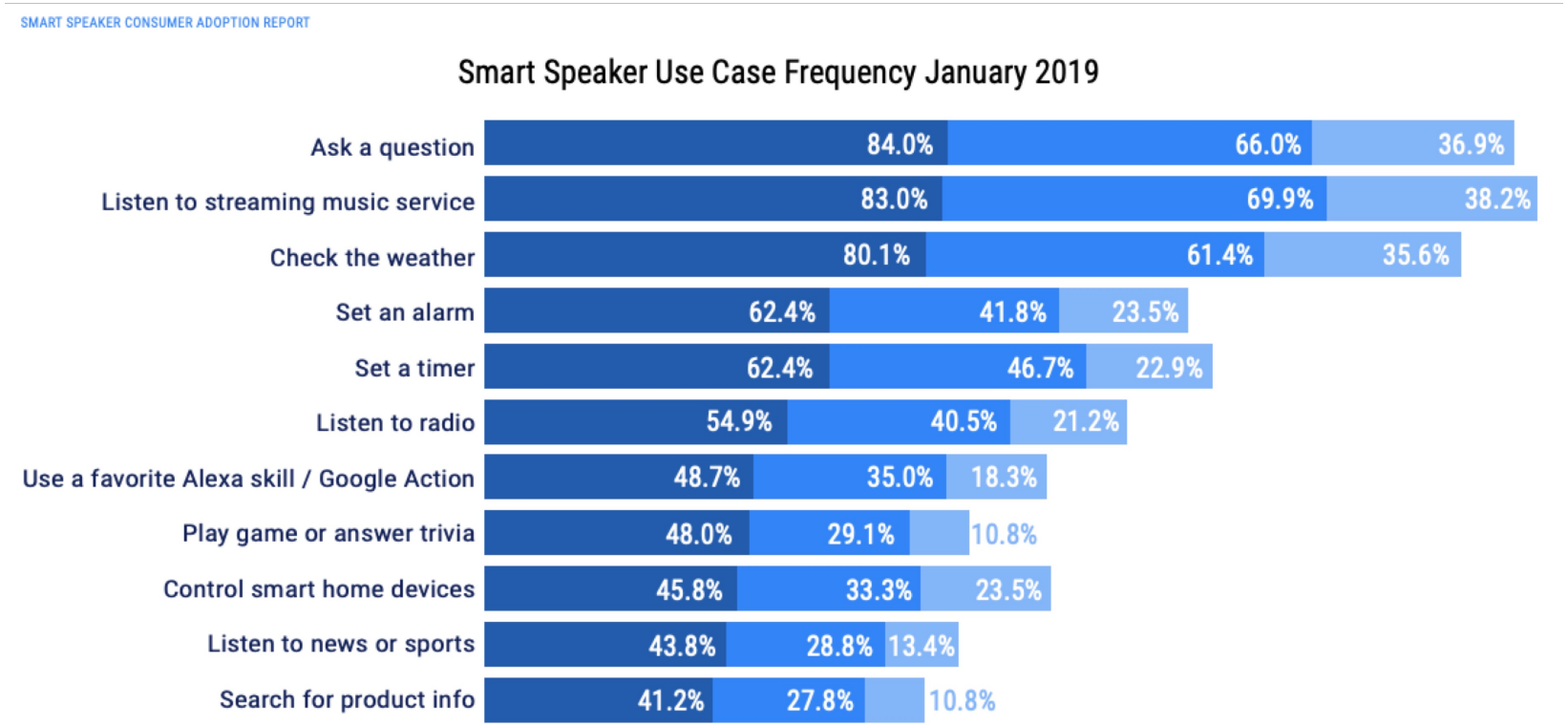
what to do over christmas in Edinburgh?

To spend Christmas in Edinburgh, you can **visit the markets, go ice skating, and attend seasonal concerts and Hogmanay events**. The festival atmosphere runs from mid-November to early January, with activities centered around Princes Street Gardens and George Street.

Christmas markets and rides

- **East Princes Street Gardens Christmas Market:** This traditional Christmas market features around 70 stalls from Scottish and European artisans, along with food and drink vendors.
- **The Big Wheel:** Located in East Princes Street Gardens, this 46m-tall Ferris wheel offers panoramic views of the city.
- **West Princes Street Gardens Family Funfair:** A free-to-enter area with funfair rides and attractions for families. It also features food stalls, warming zones, and live entertainment at the Ross Bandstand.

Applications



Why work on QA?

QA is a great test bed for evaluating how well computer systems "understand" human language

Many other NLP tasks can be reduced to QA:

Machine Translation:

Question: How do you say the following sentence in Spanish?

Paragraph: The quick brown fox jumps over the lazy dog.

Why work on QA?

QA is a great test bed for evaluating how well computer systems "understand" human language

Many other NLP tasks can be reduced to QA:

Information Extraction: (Barack Obama, educated_at, ?)

Question: where did Barack Obama graduate from?

Paragraph: Obama was born in Honolulu, Hawaii. After graduating from Columbia University in 1983, he worked as a community organiser in Chicago.

Why work on QA?

QA is a great test bed for evaluating how well computer systems "understand" human language

Many other NLP tasks can be reduced to QA:

Part-of-Speech Tagging:

Question: What is the part of speech of [runs] in the sentence?

Paragraph: He runs fast in the morning.

Why work on QA?

QA is a great test bed for evaluating how well computer systems "understand" human language

Many other NLP tasks can be reduced to QA:

Math Word Problems:

Question: What is the solution to the following problem?

Paragraph: Lisa has 7 apples. She buys 12 more apples at the grocery store. Then, she gives 5 apples to her friend. How many apples does Lisa have now?

Why work on QA?

QA is a great test bed for evaluating how well computer systems "understand" human language

Many other NLP tasks can be reduced to QA:

Language Modeling:

Question: What is the next word in the following sentence?

Paragraph: Despite the heavy rain, the match continued without any

Reading Comprehension

Reading Comprehension

Comprehend a passage of text, and answer questions about its content.

$$(P, Q) \rightarrow A$$

Where:

P is the Paragraph

Q is the Question

A is the Answer

Extractive QA: extract answer word for word from context

Reading Comprehension

Beyoncé Giselle Knowles-Carter is an American singer, songwriter, record producer and actress. Born and raised in Houston, Texas, she performed in various singing and dancing competitions as a child, and rose to fame in the late 1990s as lead singer of R&B girl-group Destiny's Child. Managed by her father, Mathew Knowles, the group became one of the world's best-selling girl groups of all time. Their hiatus saw the release of Beyoncé's debut album, Dangerously in Love (2003), which established her as a solo artist worldwide, earned five Grammy Awards and featured the Billboard Hot 100 number-one singles "Crazy in Love" and "Baby Boy".

Question: When did Beyonce start becoming popular?

Answer: in the late 1990s

Reading Comprehension

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Question: In what areas did Beyonce compete in when she was young?

Answer: singing and dancing

Reading Comprehension

Beyoncé Giselle Knowles-Carter is an American singer, songwriter, record producer and actress. Born and raised in Houston, Texas, she performed in various singing and dancing competitions as a child, and rose to fame in the late 1990s as lead singer of R&B girl-group Destiny's Child. Managed by her father, Mathew Knowles, the group became one of the world's best-selling girl groups of all time. Their hiatus saw the release of Beyoncé's debut album, Dangerously in Love (2003), which established her as a solo artist worldwide, earned five Grammy Awards and featured the Billboard Hot 100 number-one singles "Crazy in Love" and "Baby Boy".

Question: In what city and state did Beyonce grow up?

Answer: Houston, Texas

Stanford Question Answering Dataset (SQuAD)

- 100k annotated *passage-question-answer* triples
- Large-scale supervised datasets were instrumental for training effective neural RC models
- Passages are selected from the English Wikipedia
- Crowdsourced questions
- Each answer is a short segment of text (*span*) in the passage.
- It does not include questions where the answer are not mentioned in the span, and *unanswerable questions*

SQuAD was very popular (2016-2018) — today is considered "almost solved" since neural models exceed human performance

In meteorology, precipitation is any product of the condensation of atmospheric water vapor that falls under **gravity**. The main forms of precipitation include drizzle, rain, sleet, snow, **graupel** and hail... Precipitation forms as smaller droplets coalesce via collision with other rain drops or ice crystals **within a cloud**. Short, intense periods of rain in scattered locations are called "showers".

What causes precipitation to fall?
gravity

What is another main form of precipitation besides drizzle, rain, snow, sleet and hail?
graupel

Where do water droplets collide with ice crystals to form precipitation?
within a cloud

Evaluation

Metrics: **Exact Match** (EM; 0 or 1) and **F1** (partial credit)

- Multiple valid answers for each question
- Take the score which is the maximum across alternatives
- Normalisation: remove articles (a, an, the) and punctuation
- We then average EM and F1 over the whole dataset.

Q: What did Tesla do in December 1878?

A: {left Graz, left Graz and severed all relations with his family}

Prediction: *left Graz and severed*

EM: max{0, 0} = 0

Precision: 2/4, 4/4
Recall: 2/2, 2/9

F1: max{0.67, 0.61} = 0.67

Neural Reading Comprehension

Problem:

Input:

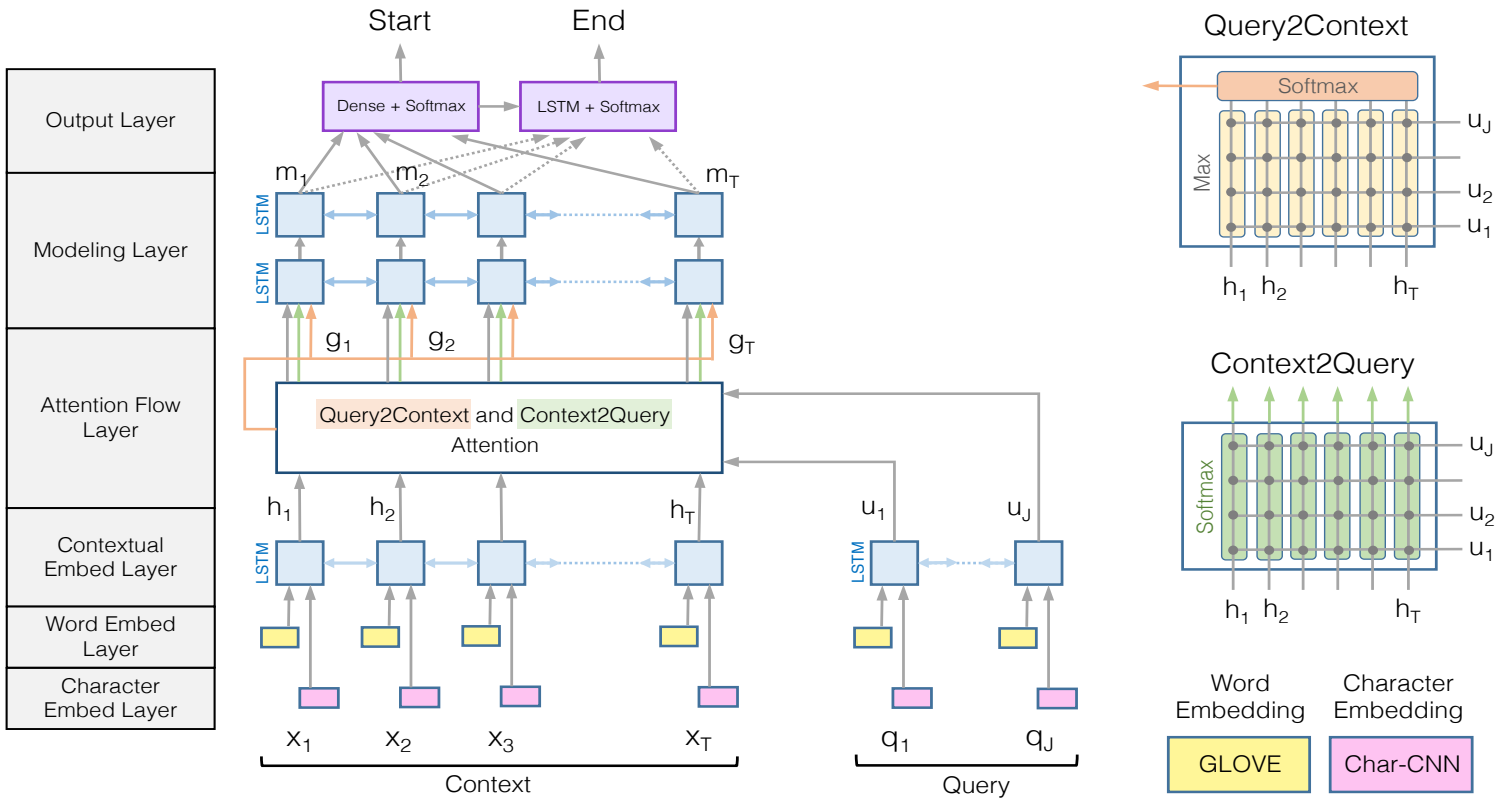
context/paragraph $C =< c_1, \cdots, c_n >$, question $\mathrm{Q} =< q_1, \cdots, q_m >, c_i, q_j \in V$

Output:

$1 \leq answer\ start\ index \leq answer\ end\ index \leq n$

(start and end indices in the provided context/passage)

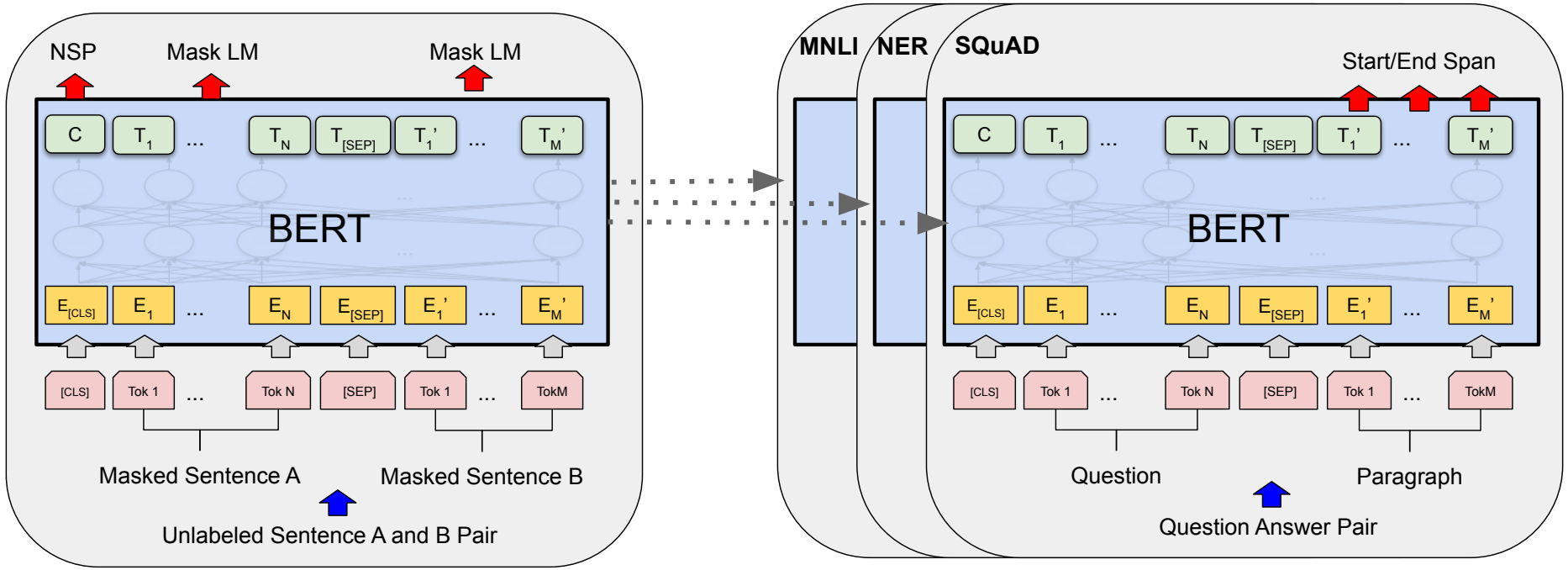
BiDAF



~2.5M params
Several BiLSTMs
Trained from scratch (minus GloVe)

Seo et al. (2016) Bi-Directional Attention Flow

BERT-based Models



Pre-training

Fine-Tuning

340M params
Transformer encoder (**no recurrence**)
Pre-Trained

Devlin et al. (2019) BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding, ACL

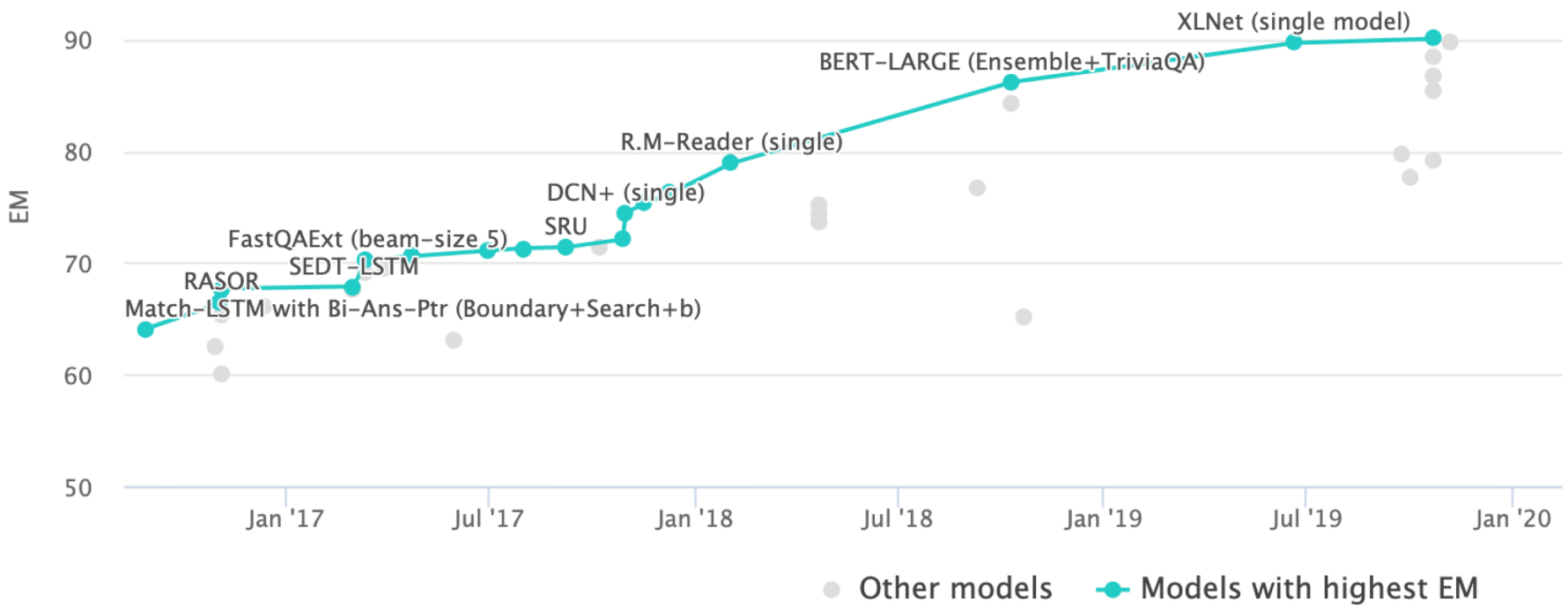
BiDAF vs BERT-based Models

	EM	F1
Human performance	82.3	91.2
BiDAF	-	85.8
BERT-large	85.1	91.8

Results: Close to human performance (almost) without any architecture engineering/tweaking!

Devlin et al. (2019) BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding, ACL

Stanford Question Answering Dataset (SQuAD)



https://rajpurkar.github.io/SQuAD-explorer/

Why look at Reading Comprehension?

- It bridges foundational NLP and modern LLM approaches
  - Old school NLP: feature engineering, bespoke architectures for tasks
  - Modern deep learning: transformers, pretraining and fine-tuning

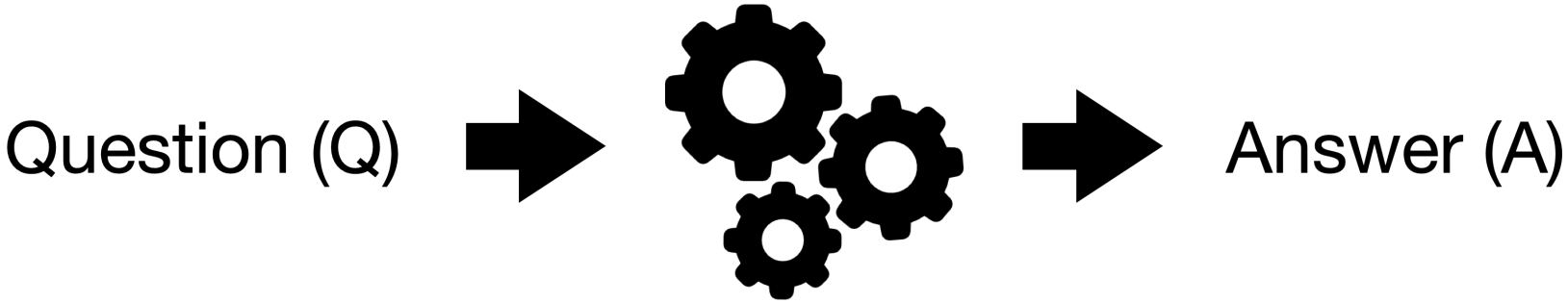
- Reading comprehension simple precursor to many current research:
  - Multi-hop QA and reasoning
  - Retrieval-augmented QA

- Many real world application are advanced reading comprehension:
  - Document QA for legal, scientific, medical
  - Chatbots over policies/products
  - Assistants that analyse reports

Most current usage of LLM is a variant of QA

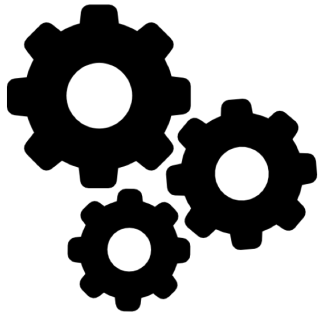
Open Domain QA

Open Domain QA



Knowledge is also encoded in large language models, but can we do better when knowledge is retrieved from a trusted knowledge source?

Open Domain QA

Question (Q) →  → Answer (A)

Retrieval from Knowledge Source

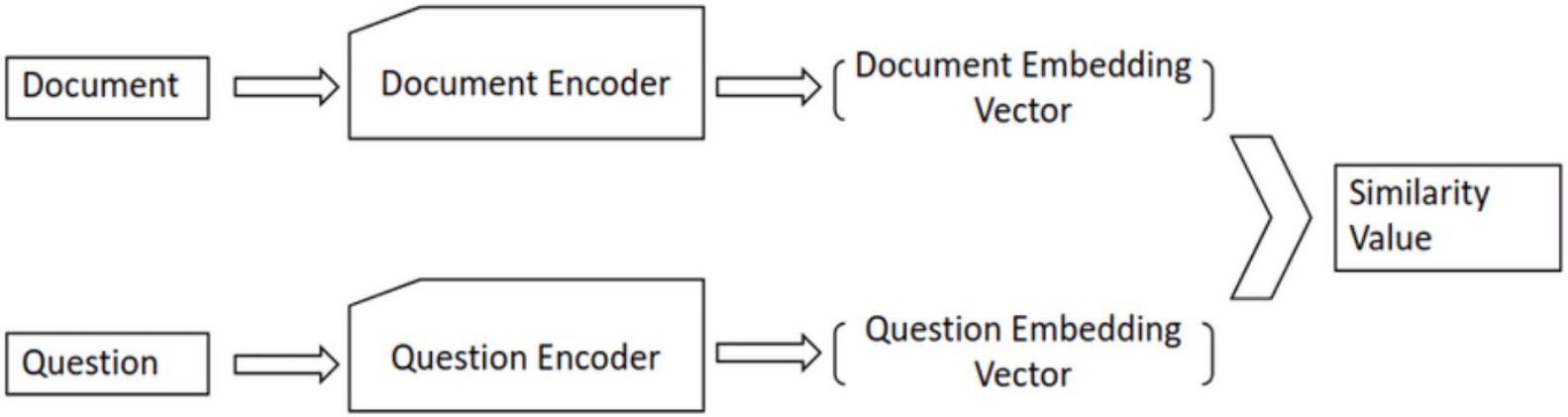
$(P_1, \cdots, P_k)$

- **Open-Domain Question Answering (ODQA):** Not given one small passage
- We can access a large collection of documents (e.g., Wikipedia) — we don't know which document contains the answer, and the goal is to answer any open-domain questions.
- Both more challenging and more practical/useful!

Dense Passage Retrieval

Information retrieval: search for passages relevant to question

1) Embed “Passage/Document” and “Question” as dense vectors (eg. using BERT encoder)
2) Then find **similarity** by taking the **dot product** between them
3) Select top k passages using **approximate nearest neighbours** (More details in Lecture 14 on RAG)



Natural Questions Dataset

- First large dataset of **genuine Google Search** queries, paired with evidence from Wikipedia.
- Entire Wikipedia pages to handle (up to 5k tokens) - full documents not short passages
- ODQA systems had to shift from toy questions to *real-world, noisy, multi-word, ambiguous* queries *over full documents* not short passages
- Led to the development of RAG and Longer context LLMs

Natural Questions Annotation

Question: when was the egg mcmuffin added to the menu (Real question from Google)

Wikipedia page:

McMuffin

11 languages

Article Talk Read View source View history Tools

From Wikipedia, the free encyclopedia

McMuffin is a family of breakfast sandwiches sold by the international fast food restaurant chain McDonald's. The **Egg McMuffin** is the signature sandwich, which was invented in 1972 by Herb Peterson to resemble eggs benedict, a traditional American breakfast dish with English muffins, ham, eggs and hollandaise sauce.[1]

Product description

In the US and Canada the standard McMuffin consists of a slice of Canadian bacon,[2] a griddle-fried egg, and a slice of American cheese on a toasted and buttered English muffin. The round shape of the egg is made by cooking it in a white plastic ring surrounded by an outer metal structure.[3][1]

History

The sandwich was invented in 1972.[4] Former McDonald's President Ray Kroc wrote that Herb Peterson and his assistant, Donald Greadel, the operator of a McDonald's Santa Barbara franchise in Goleta, California,[5] asked Kroc to look at something, without giving details because it was:

> ... a crazy idea — a breakfast sandwich. It consisted of an egg that had been formed in a Teflon circle with the yolk broken, and was dressed with a slice of cheese and a slice of grilled ham. It was served open-faced on a toasted and buttered English muffin. The advent of the Egg McMuffin opened up a whole new area of potential business for McDonald's, the breakfast trade.[3][1]

McMuffin

Nutritional value per 1 sandwich, 7.1 oz (200 g)

Energy	300 kcal (1,300 kJ)	
Carbohydrates	30 g (10%)	
Sugars	16 g	
Dietary fiber	2 g (8%)	
Fat	18 g (37%)	
Saturated	5 g (24%)	
Trans	0†	
Protein	18 g	
Vitamins	**Quantity**	**%DV†**
Vitamin A equiv.	90 µg	11%
Vitamin C	0 mg	0%
Vitamin E	0 mg	0%

Step 1: Annotator selects context

The first McDonald's Corporate-authorized Egg McMuffin was served at the Belleville, New Jersey McDonald's in 1972.

Step 2: Annotator selects short answer, where applicable.

Kwiatkowski et al. (2019). Natural Questions: A Benchmark for Question Answering Research TACL

Natural Questions Challenges

Question:

what does the word china mean in chinese?

Short Answer:

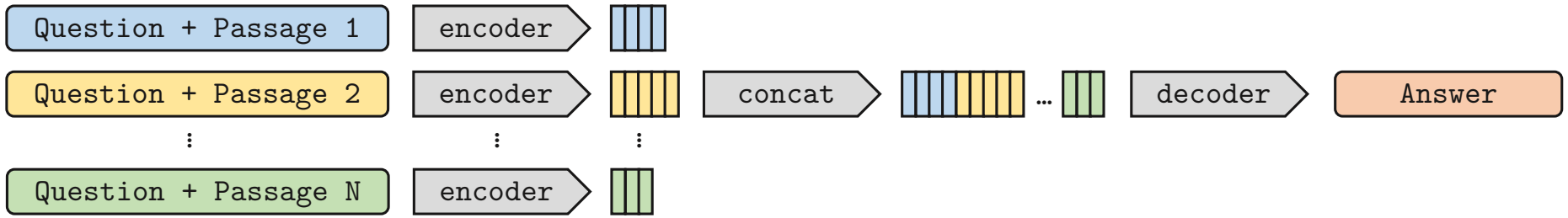
None

Long Answer:

The names of China include the many contemporary and historical appellations given in various languages for the East Asian country known as Zhongguo (中國 / 中国) in its official language . China , the name in English for the country , was derived from Portuguese in the 16th century , and became popular in the mid 19th century . It is believed to be a borrowing from Middle Persian , and some have traced it further back to Sanskrit . It is also generally thought that the state of Qin that later formed the Qin dynasty is the ultimate source of the name , although there are other suggestions.

- **Unanswerable**: some have no short answer, and some have no answer at all
- **Ambiguous** questions and **long-form answers**

Fusion in Decoder



- Combined advantages of extractive reading comprehension models, with advantages of generative QA systems like T5 which relied only on parameters: retrieve and generate!
- Question+Passage are processed independently in the encoder (T5 model) with added tokens “question:”, “title:”, “context:”
- Decoder attends over the concatenation of all retrieved passages: fusion in the decoder.
- Can better aggregate evidence from multiple passages

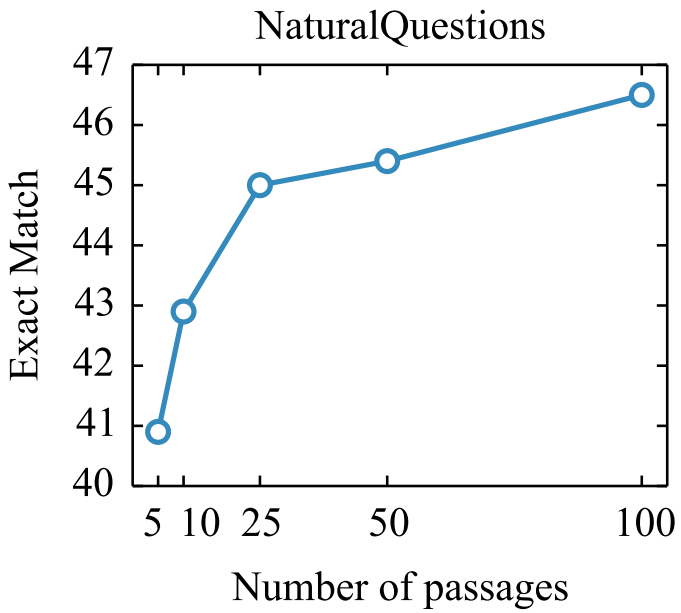
Izacard and Grave (2021). Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering, EACL

Fusion in Decoder (FiD)

Exact match: discard answers > 5 words, normalize lowercase, remove punctuation & articles

Q: what could problems be with this evaluation?

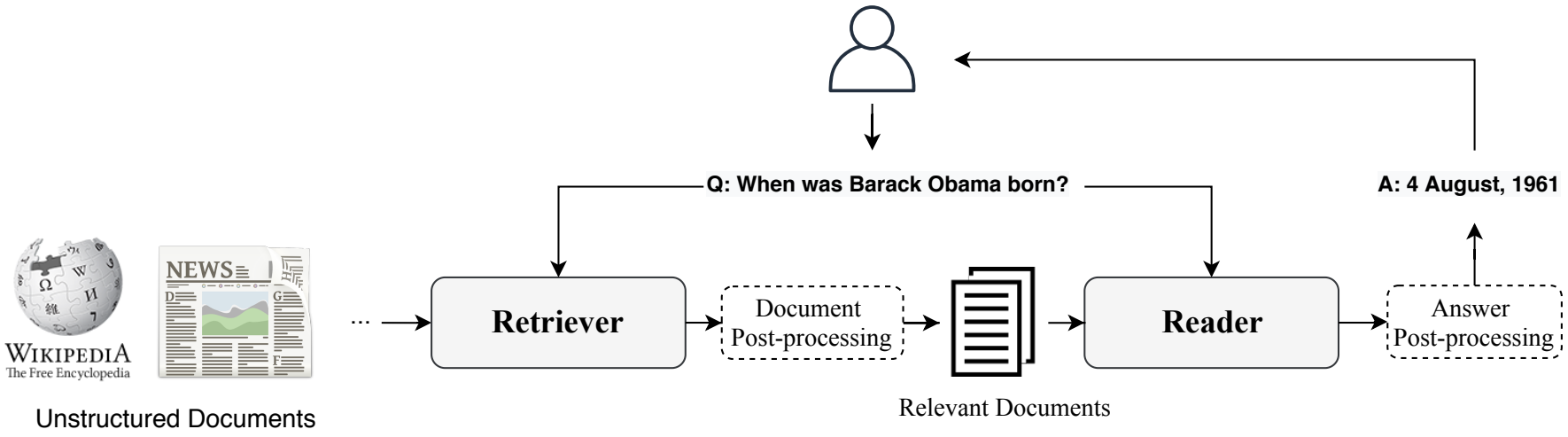
T5 (Roberts et al., 2020)	36.6	11B
GPT-3 few shot (Brown et al., 2020)	29.9	175B
Fusion-in-Decoder (base)	48.2	220M
Fusion-in-Decoder (large)	**51.4**	770M



Foundation for modern RAG and follow-up work

Izacard and Grave (2021). Leveraging Passage Retrieval with Generative Models for Open Domain Question Answering, EACL

Retriever-Reader Architecture



- Retrieval Augmented Generation – Lecture 14 in week 5
- Long Context modelling – Lecture 11 in week 4

Figure from Zhu et al. (2021). Retrieving and Reading : A Comprehensive Survey on Open-domain Question Answering

Summary

- QA is related to all other tasks: translation, reasoning etc.
- For this lecture limited to knowledge retrieval questions
- Early Machine Reading Comprehension (MRC) methods are defined by a question, and a passage of text and finding an answer in from that passage
- Stanford Question Answering Dataset (SQuAD) established MRC as a flagship NLP task - large scale driving progress in field
- Open Domain QA introduced the idea of retrieval
- Natural Questions led to large corpus of real questions with short and long answers and unanswerable questions.
- Fusion in Decoder showed QA could combine retrieval with generation